

CA2 MPC: Assignment Instructions

1. Implement a Linear MPC Controller with Constraints

Modify the function `xy_axis_mpc.m` to implement a linear Model Predictive Controller (MPC) with constraints. The constraints for the x-y velocity, acceleration, and jerk are specified in **Slide 57 of Lecture 5**. Ensure that these constraints are properly incorporated into the optimization problem.

2. Construct the Required Matrices

You will need to construct the following matrices: **H**, **F**, **A**, and **b**. These matrices will be passed as inputs to the `quadprog` function to solve the optimization problem. Pay attention to how these matrices are formed, as they define the dynamics and constraints of the MPC.

3. Simulate the Trajectory

After completing the implementation, run the file `home_work.m` to visualize the simulated trajectory. Ensure that the resulting trajectory meets the defined constraints and behaves as expected.

4. Review Helpful Resources

To assist with your implementation, you are encouraged to review the examples provided in the files:

- `linear_mpc.m`
- `linear_mpc_hard_constraint.m`
- `linear_mpc_soft_constraint.m`

These files contain useful references for constructing the MPC controller with both hard and soft constraints.